Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Convex Perturbation Paper - My Response to Your Latest Comments

Benjamin D. Summers <bsummers@adagiollc.com> Thu, Sep 24, 2026 at 2:44 AM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Hi Ezequiel and Sara,

Sara, noted that you have no further points; thank you.

Ezequiel, thank you for the verification record and for the two results; I adopted both, and the four wording points. The revised manuscript is on SSRN under the same DOI (https://dx.doi.org/10.2139/ssrn.6812461):

Attainment (your item 1). Remark 4.5 now states that if a triple attains both suprema with K = −cX and c = M_/D_CK, then S(g) = D_CK + gM_, the deficit equals E(g) on [0,1], and g_first = g_term = g_1; hence no bound in D_CK and M_* alone improves Theorem 4.3. Section 4.3 realizes the configuration: K = 2(q − 0.3)(0.2 − q)Z with q = ½(1 − e^{−2λ}), aligned against X for λ ∈ (0.2554, 0.4581); at λ = 0.35 the response equals E to machine precision and g_first = g_term = g_1 = 0.5311 at ε = 0.04, against g_0 = 0.5555. The abstract and the contributions paragraph state the attainment.

Scale-level family (your item 2), in place of the extension discussed in item 10. Proposition 4.10, new: Γ_{g*} is divisible for an interior g* if and only if Γ_0 = (Γ_a − gΓ_ref)/(1 − g) for a divisible Γ_a with Γ_a ≥ gΓ_ref entrywise; then X = gK at every triple and D_CK(δt;Γ_g) = (1 − g)|g* − g| M_(δt). The instance is your Glauber pair (reference λ = 2, extrapolant λ = 1, g = 0.3603), with the constants in closed form: with a(τ) = (e^{−2τ} − e^{−4τ})/(1 − g*), M_(δt) = a(½ ln 2)² = 1/[16(1 − g)²] = 0.152731 for δt ≥ ln 2, D_CK(δt;Γ_0) = gM_ = 0.055029, and the rebound peak is (1 − g*)²M_/4 = 1/64 = 0.015625 exactly, at g = (1 + g)/2. At ε = 0.01: g_first = 0.2705, g_term = 0.8721, g_1 = 0.9500, g_0 = 0.9519, as you report. One figure differs: the re-entry point from the closed form is 0.4882, not 0.4885 (g² − (1 + g*)g + g* + ε/M_* = 0 has roots 0.4882 and 0.8721), and the manuscript prints 0.4882. The abstract now says that a two-state family exhibits the interior zero, the rebound, and the threshold separation at the scale level; the reference to the Koopman paper's Section 6 family is removed. Sara, the pointwise example keeps the status statement you asked for; the scale-level family sits after it.

Wording. (i) Abstract: "exact vanishing is guaranteed at g = 1 and possible at no more than one interior coupling." (ii) After Proposition 6.1: "reduces to the bound (7) of Theorem 3.1." (iii) Remark 6.2 states the weight for states of positive probability under the mixture; a null column is completed with the constant g, which keeps the family in class (16) and leaves δ_g and the estimate unchanged, because both one-time laws sum to one over the states of positive probability, so g lies between the smallest and the largest Bayes weight at each s. (iv) Proof of Theorem 4.3(i): the radicand is positive because (D + M)² − 4Mε > (D + M)² − 4MD = (D − M)² ≥ 0 for ε < D.

Proposition 4.10 is the only new numbered item; every other number is unchanged, so the references in both of your letters still hold.

Let me know your thoughts.

Ben

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On Wed, Sep 23, 2026 at 5:34 PM Ezequiel Boero <ezequiel.boero@adagioinstitute.org> wrote:

> Ben,
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> I'm fulfilling here my reply to your responses on paper 3.
> So far I could see, Sara and I raised three principal issues; all three are addressed.